Behzad Karimi

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EDUCATION

Montana State University Jan 2024 – May 2026
M.S. in Electrical & Computer Engineering
GPA: 3.80/4.00; Adviser: Prof. Bradley Whitaker
Thesis: Machine Vision–Based Automated Flake Detection with Feedback-Driven Model Improvement
Coursework: Machine Learning, Mathematics of Machine Learning, Advanced Data Mining, Data Visualization, Methods of Data Analysis, Machine Learning for Engineering

Ghiaseddin Jamshid Kashani University, Iran 2009
B.S. in Computer Engineering

RESEARCH INTERESTS

Machine Learning • Computer Vision • Deep Learning • Generative AI • Data-Centric AI • Scientific Machine Learning

PUBLICATIONS

1. **Karimi, B**., & Whitaker, B. M. (2026, May). Machine Vision-Based Automated Flake Detection with Feedback-Driven Model Improvement. In 2026 Intermountain Engineering, Technology and Computing (IETC) (pp. 1-6). IEEE.

2. Chegini, T. G., Shivanian, E., **Karimi, B**., & Dadgostari, F. (2026). Sparse-Observation Atmospheric Thermal Forecasting with Physics-Informed Neural Networks for Climate-Aware Digital Twins. arXiv preprint arXiv:2609.27290.

RESEARCH EXPERIENCE

Research Assistant, MonArk NSF Quantum Foundry, Montana State University Jan 2024 – Aug 2025

- Contributed to the development of an automated machine learning pipeline for flake detection in two-dimensional quantum materials.
- Implemented data preprocessing and organization strategies, including sorting detected flakes by label and area to prioritize thin flakes critical for quantum research.
- Developed tools to convert detection outputs from CSV format to COCO-style JSON annotations, enabling automated model retraining.
- Supported the integration of the pipeline across evolving software platforms, adapting code to updated syntax, dependencies, and system configurations.
- Implemented a feedback loop to support iterative retraining as new labeled data became available, while identifying potential bias introduced by self-labeled data.
- Worked with PyTorch-based computer vision models and transfer learning approaches (e.g., DetectoRS) to improve detection speed and scalability.

TEACHING EXPERIENCE

Graduate Teaching Assistant, Signals and Systems, Montana State University Aug 2025 – May 2026
Led lab sessions, supported students, and helped grade and prepare lab and course materials.

Instructor, Tehran Institute of Technology, Iran 2012 – 2018
Taught Python programming, networking fundamentals (Network+), and ICDL computer skills.

PROFESSIONAL EXPERIENCE

IT Administrator, Padzanij, Iran 2017 – 2022
Designed and maintained the company's network and server infrastructure (Windows Server, Active Directory, DNS, DHCP, RAID) and managed IT and accounting systems, including backups.

TECHNICAL SKILLS

Programming: Python, C++, SQL, R, MATLAB
ML & Vision: PyTorch, TensorFlow, Scikit-learn, OpenCV, MMDetection, object detection, Reinforcement Learning
Data & Tools: NumPy, pandas, Matplotlib, Git, Docker, Conda, LaTeX, Tableau

CERTIFICATES

Machine Learning Specialization (Coursera): Supervised Machine Learning: Regression and Classification (2024); Advanced Learning Algorithms (2026)